

Translated

From: e-maxx.ru

D'Esopo-Pape algorithm

Given a graph with n vertices and m edges with weights w_i and a starting vertex v_0 . The task is to find the shortest path from the vertex v_0 to every other vertex.

The algorithm from D'Esopo-Pape will work faster than [Dijkstra's algorithm](#) and the [Bellman-Ford algorithm](#) in most cases, and will also work for negative edges. However not for negative cycles.

Description

Let the array d contain the shortest path lengths, i.e. d_i is the current length of the shortest path from the vertex v_0 to the vertex i . Initially this array is filled with infinity for every vertex, except $d_{v_0} = 0$. After the algorithm finishes, this array will contain the shortest distances.

Let the array p contain the current ancestors, i.e. p_i is the direct ancestor of the vertex i on the current shortest path from v_0 to i . Just like the array d , the array p changes gradually during the algorithm and at the end takes its final values.

Now to the algorithm. At each step three sets of vertices are maintained:

- M_0 - vertices, for which the distance has already been calculated (although it might not be the final distance)
- M_1 - vertices, for which the distance currently is calculated
- M_2 - vertices, for which the distance has not yet been calculated

The vertices in the set M_1 are stored in a bidirectional queue (deque).

At each step of the algorithm we take a vertex from the set M_1 (from the front of the queue). Let u be the selected vertex. We put this vertex u into the set M_0 . Then we iterate over all edges coming out of this vertex. Let v be the second end of the current edge, and w its weight.

- If v belongs to M_2 , then v is inserted into the set M_1 by inserting it at the back of the queue. d_v is set to $d_u + w$.